Explainability-Guided SegFormer for Forest Cover Segmentation Using Attention Consistency Supervision

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Abstract

Vision Transformer (ViT)-based segmentation models achieve strong accuracy, but their internal attention is typically used only for post-hoc visualization, not as a supervised training signal. Prior work applying attention supervision for interpretability has focused on medical imaging and autonomous-driving domains, while forest/non-forest segmentation from aerial imagery still treats attention purely as an architectural feature-refinement mechanism, with interpretability evaluated qualitatively if at all. We present an explainability-guided SegFormer-B0 architecture that treats attention as a first-class training target: an *Attention Consistency Loss* aligns the model's internal attention with the ground-truth forest mask during training, encouraging the network to focus on canopy regions rather than roads, shadows, or built structures. Because SegFormer's encoder uses spatial-reduction attention rather than standard multi-head self-attention, we adapt Gradient-weighted Attention Rollout to SegFormer, restricting it to the final encoder stage where attention is square and token-consistent. We introduce a quantitative metric, Average Attention–Mask Overlap (AAMO), to evaluate interpretability numerically rather than only visually. On the full 5,108-image dataset (3576/766/766 split, seed 42, 20 epochs), with λ_2 set by a validation sweep, the proposed method raises AAMO from 0.0334 to 0.7476 (~22×) relative to a vanilla SegFormer-B0 baseline trained on the same split, at a modest segmentation cost (Dice 0.8743→0.8577, IoU 0.7766→0.7508), alongside a full-scale-trained U-Net baseline (Dice 0.8615, IoU 0.7568) on the same held-out test set. A complete multi-seed ablation study is ongoing.

CCS Concepts

• **Computing methodologies → Neural networks**; **Image segmentation**; *Interpretability and explainability*.

Keywords

semantic segmentation, vision transformers, SegFormer, attention supervision, explainable AI, forest cover mapping, remote sensing

1 Introduction

Semantic segmentation of forest cover from aerial imagery underpins ecological monitoring, deforestation tracking, and land-use management. Transformer-based models such as SegFormer [9] have been applied to this task for their ability to capture long-range spatial dependencies, but the great majority of transformer segmentation work, including recent forest-specific studies [1, 5, 7], uses attention purely as an architectural mechanism to improve accuracy, evaluating explainability only through post-hoc qualitative heatmap inspection, if at all.

Transformer attention in segmentation models is not explicitly guided during training and frequently attends to task-irrelevant regions — roads, shadows, buildings — rather than the class of interest, reducing both accuracy at ambiguous boundaries and the trustworthiness of the model's explanations. No existing work addresses this specifically for forest/non-forest segmentation, nor evaluates attention faithfulness quantitatively in this domain.

This paper makes four contributions: (1) an Attention Consistency Loss that supervises SegFormer-B0's internal attention against the ground-truth forest mask during training; (2) an adaptation of Gradient-weighted Attention Rollout to SegFormer's spatial-reduction attention; (3) AAMO, a quantitative interpretability metric evaluable alongside standard segmentation metrics; and (4) a preliminary empirical comparison against U-Net and vanilla SegFormer-B0 baselines showing that explainability-guided training substantially improves attention faithfulness at only a modest segmentation-accuracy cost.

2 Related Work

Forest-cover segmentation from aerial and satellite imagery has been approached almost exclusively as an accuracy problem. U-Net [8] remains the dominant convolutional baseline, and recent forest-specific datasets such as the WildUAV forest-inspection benchmark [2] have focused effort on acquiring larger and more varied labeled imagery (real and simulated, multiple altitudes and illumination conditions) to improve segmentation accuracy on models such as HRNet and PointFlow, without examining what the underlying network attends to or why. Existing work directly relevant to attention supervision falls into two families.

Attention-as-architecture. ForResANeXt [5] embeds lightweight attention in residual blocks for forest/non-forest satellite segmentation; Attention-Refined PP-LiteSeg [7] fuses multi-branch attention for cross-regional UAV forest segmentation; an attention-based U-Net [1] adds attention gates for deforestation mapping on Sentinel-2 imagery. All three use attention purely as a feature-refinement mechanism improving accuracy — attention is never supervised or evaluated as an explanation.

Attention-as-explainability. eX-ViT [10] regularizes ViT attention with an attribute-guided loss for weakly supervised natural-image classification; TransAttUnet [4] integrates guided attention into a U-Net/Transformer hybrid for medical segmentation; work on driving models [6] applies an explicit attention loss to align

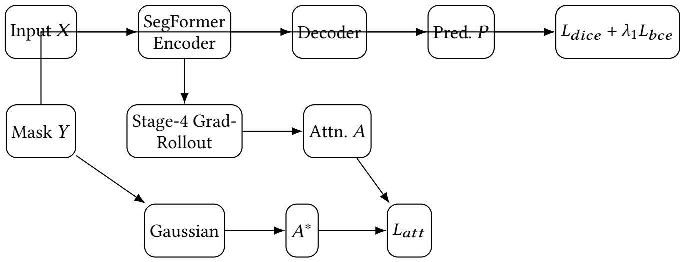

Figure 1: Training pipeline: predicted mask P and attention map A are each supervised, then combined into the total training objective L.

Algorithm 1 One training step of the attention-consistency variant

Require: image X, ground-truth mask Y, model (E, D), weights λ_1, λ_2

1: $F \leftarrow E(X)$ ▷ encoder features, all 4 stages
2: $P \leftarrow D(F)$ ▷ decoder segmentation output
3: $A \leftarrow \text{Rollout}(F_4)$ ▷ stage-4 attention only, autograd-attached
4: $A^* \leftarrow \text{Gaussian}(Y, \sigma{=}8)$ ▷ soft attention target
5: $L \leftarrow L_{dice}(P, Y) + \lambda_1 L_{bce}(P, Y) + \lambda_2 L_{att}(A, A^*)$
6: L.backward(create_graph=True) ▷ double backprop: A depends on ∇F
7: optimizer.step()

attention with task-relevant scene regions. All three supervise attention as a training signal, but outside the forest/remote-sensing domain and without a quantitative attention-mask overlap metric. The attention-extraction method underlying our approach is Chefer et al.'s Gradient-weighted Attention Rollout [3], originally defined for standard multi-head self-attention.

Gap. No prior work combines (a) attention supervision as a training signal, (b) the forest/non-forest remote-sensing domain, and (c) a quantitative interpretability metric evaluated alongside standard segmentation metrics. The attention-as-architecture literature [1, 5, 7] operates in the correct domain but never supervises or measures attention as an explanation. The attention-as-explainability literature [4, 6, 10] supervises attention as a training signal but never in the forest-segmentation domain, and none of the three reports a quantitative attention-fidelity metric comparable across model configurations. This paper addresses that combined gap: an Attention Consistency Loss that supervises SegFormer's internal attention against the ground-truth forest mask, an adaptation of Gradient-weighted Attention Rollout to SegFormer's spatial-reduction attention, and AAMO, a new quantitative metric that lets attention fidelity be reported and compared the same way Dice or IoU are. We also draw on the WildUAV forest inspection dataset work [2] as evidence that forest-aerial-segmentation research so far has focused effort on acquiring larger and more varied labeled imagery rather than on what the resulting models attend to.

3 Proposed Method

3.1 Architecture Overview

The pipeline (Figure 1) extends SegFormer-B0 with an explainability-supervision path applied during training only; at inference the model runs as standard SegFormer-B0 with negligible overhead. The encoder's stage-4 attention (the only stage with spatial-reduction ratio = 1, giving square, token-consistent 64×64 self-attention) feeds an adapted Gradient-weighted Attention Rollout to produce attention map A. The ground-truth mask Y is Gaussian-smoothed into a soft target A^*. A and A^* are compared by the Attention Consistency Loss and combined with the standard segmentation loss.

3.2 Handling Spatial-Reduction Attention

Standard Attention Rollout assumes full multi-head self-attention across all tokens. SegFormer's Efficient Self-Attention reduces keys/values by a spatial-reduction ratio at earlier stages (sr = 8, 4, 2 for stages 1–3), making those stages' attention matrices rectangular rather than square. Only stage 4 (sr = 1) has true square, token-consistent 64×64 attention — the only stage where Rollout's recursive matrix product is valid without further approximation, verified empirically on MiT-B0 at 256×256 input resolution.

3.3 Mathematical Formulation

Let $X \in \mathbb{R}^{256\times256\times3}$ and ground-truth mask $Y \in \{0,1\}^{256\times256}$:

$$F = E(X), \quad P = D(F) \in [0,1]^{256\times256} \tag{1}$$

$$A = \text{Rollout}(F) \in [0,1]^{256\times256} \tag{2}$$

$$A^* = \text{Gaussian}(Y) \tag{3}$$

$$L_{dice} = 1 - \frac{2|P \cap Y|}{|P| + |Y|}, \quad L_{bce} = -\sum \Big[Y \log P + (1-Y)\log(1-P) \Big] \tag{4}$$

$$L_{att} = \text{MSE}(A, A^*) \text{ or } \text{KL}(A \,\|\, A^*) \tag{5}$$

$$L = L_{dice} + \lambda_1 L_{bce} + \lambda_2 L_{att} \tag{6}$$

with λ_1=1 and λ_2 tuned on the validation set: a sweep over $\lambda_2 \in \{0.1, 0.3, 0.5, 1.0\}$ (MSE, all else fixed) selects λ_2=1.0, which maximizes test AAMO. Because A is itself defined via a gradient (Grad-Rollout is Grad-CAM-style), backpropagating L_{att} requires double backpropagation; our implementation keeps A attached to the autograd graph via `create_graph=True` so one combined `backward()` reaches all parameters.

Algorithm 1 summarizes one training step. At inference, only lines 1–2 run — the Rollout/attention path is training-only supervision, so deployment cost equals vanilla SegFormer-B0.

3.4 AAMO: Average Attention–Mask Overlap

Standard practice evaluates attention maps only qualitatively, by visual inspection of heatmaps. We instead define AAMO to make attention fidelity reportable and comparable the same way Dice or IoU are. Given normalized attention $A \in [0,1]^{256\times256}$ and ground-truth mask Y, AAMO is the fraction of forest pixels covered by thresholded attention:

$$A_{thr} = \mathbb{1}[A > \tau], \quad \tau = 0.5 \tag{7}$$

$$\text{AAMO} = \frac{|A_{thr} \cap Y|}{|Y|} \tag{8}$$

i.e. AAMO is the recall of the ground-truth forest region under thresholded attention: 1.0 means attention fully covers the forest mask, 0.0 means no overlap. We additionally report $\text{AAMO}_{\text{dice}}$ and AAMO_{IoU} (Dice and IoU between A_{thr} and Y) as symmetric overlap checks, since AAMO alone does not penalize attention that also covers large non-forest regions.

4 Experimental Setup

Dataset. 5,108 RGB aerial images (256×256) with binary forest/non-forest masks; 70/15/15 stratified train/val/test split. A full dataset audit confirms 0 orphaned image/mask pairs, 0 unreadable files, and that masks are already effectively binary — no pixel sits more than 9 grey levels from pure black or white, and the mid-gray band (32–223) is empty. The dataset is moderately forest-majority (mean forest-pixel ratio 0.6117 ± 0.3306 per mask), which we account for by reporting Dice/IoU/F1 rather than pixel accuracy alone, since accuracy alone would be inflated by the majority class. U-Net and both SegFormer-B0 variants below are trained and evaluated on the full 5,108-image dataset, using an identical 3576/766/766 split (seed 42) so all three share the same 766-image held-out test set. DeepLabV3+ remains a 200-image CPU-scale smoke run, an optional extra reference point pending a full-scale run.

Baselines. U-Net (CNN) and vanilla SegFormer-B0 (no attention loss), isolating the effect of the proposed attention supervision, plus DeepLabV3+ (MobileNetV3 backbone) as an additional optional CNN reference point.

Metrics. Dice, IoU, F1 (segmentation quality); parameter count, GFLOPs, FPS (efficiency); AAMO — normalized overlap between thresholded attention and the ground-truth forest mask (interpretability).

Implementation details. SegFormer-B0 variants are trained on a GPU runtime for 20 epochs with AdamW (lr = 6×10^{-5}), batch size 16 (vanilla) / 1 (attention variant, due to the double-backpropagation memory cost), Gaussian target smoothing $\sigma = 8$, and L_{att} using the MSE formulation, seed 42. The attention-loss weight is set to $\lambda_2 = 1.0$, the value selected by the validation sweep (Section 3.3). Reported checkpoints are each variant's best-validation epoch (vanilla: epoch 5; attention-consistency: epoch 11) rather than the final epoch, since training Dice reaches ~0.90 by epoch 20 while validation Dice peaks earlier. Both the MSE and KL-divergence formulations of L_{att} are implemented and unit-tested, but only MSE has been evaluated on trained checkpoints so far.

Reproducibility. The attention-extraction and loss implementation is covered by 13 unit tests independent of any trained model (8 for the Attention Consistency Loss, 5 verifying Grad-Rollout output shapes at each SegFormer stage), and AAMO by 10 further tests on synthetic attention/mask pairs, all passing. Dataset integrity is checked by a standalone auditing script (pairing, geometry, value spread, class balance) that exits non-zero on failure, so it can run as a pre-flight gate before training rather than being assumed correct.

5 Results

Table 1 reports the full 5,108-image, 20-epoch GPU run for U-Net and both SegFormer-B0 variants (attention variant: λ_2=1.0), all on the same 3576/766/766 held-out test set. AAMO increases approximately 22× (0.0334 → 0.7476) with the attention-consistency variant, at a modest cost in Dice (0.8743→0.8577). Across the λ_2 sweep (0.1–1.0, MSE), test AAMO rises monotonically (0.548, 0.575, 0.603, 0.748) while test Dice stays within 0.852–0.869. Our earlier 400/60/60, 8-epoch CPU smoke run had instead shown the attention loss slightly *improving* Dice/IoU (0.8017⟶0.8191, 0.6690⟶0.6936); that small-scale accuracy gain did not hold at full scale, though the AAMO direction was consistent at both scales (smoke: 0.0144⟶0.432). We additionally report DeepLabV3+ (MobileNetV3 backbone) as an optional CNN reference point beyond U-Net; on a 200-image smoke run it trails both U-Net and SegFormer-B0 on Dice/IoU and shows a strong precision/recall imbalance (precision 0.59, recall 0.98), indicating a bias toward over-predicting forest rather than delineating boundaries precisely.

Table 1: Comparison on the full 5,108-image dataset (3576/766/766 split, seed 42): U-Net and SegFormer-B0 variants share this held-out test set. The attention-consistency row uses λ_2=1.0, selected by a validation sweep over $\{0.1, 0.3, 0.5, 1.0\}$; checkpoints are each variant's best-validation epoch (vanilla 5, attention-consistency 11). DeepLabV3+ remains a 200-image smoke run, an optional extra baseline not yet run at full scale.

Model	Dice	IoU	F1	AAMO
U-Net (CNN)	0.8615	0.7568	0.8615	n/a
DeepLabV3+ (MobileNetV3)	0.7369	0.5834	0.7369	n/a
SegFormer-B0	0.8743	0.7766	0.8743	0.0334
SegFormer-B0 + L_{att}	0.8577	0.7508	0.8577	**0.7476**

Augmentation ablation (U-Net, full scale). Separately from the smoke-scale numbers above, a full-scale (5,108-image, 20-epoch) augmentation ablation on the U-Net baseline — same split, seed, and starting weights, toggling only the augmentation pipeline — finds that augmentation does not improve accuracy under this training budget: test Dice falls from 0.8604 (no augmentation) to 0.8505 (with augmentation), IoU from 0.7598 to 0.7454. Neither arm overfits within 20 epochs, the regime in which augmentation is expected to pay off, and this is a single-seed result, so we read it as *augmentation did not improve the U-Net baseline under our training budget*, not as evidence that augmentation is unhelpful for this task in general.

Remaining limitations. The full 5,108-image, 20-epoch run resolves the earlier smoke-scale caveat for U-Net and both SegFormer-B0 variants, but the SegFormer checkpoints are still early-stopped at their best-validation epoch rather than fully converged, the λ_2 sweep is single-seed, and no run has yet been repeated with a different seed. DeepLabV3+ remains a 200-image smoke-scale reference only. Figure 2 shows the qualitative attention-drift comparison still does not tell as clean a story as the AAMO number: the vanilla model's attention continues to exhibit a corner/edge artifact rather than the motivating roads/shadows failure mode, and this persists at full scale, so it is unlikely to be simply an undertraining effect as earlier speculated.

6 Discussion

The large AAMO gain relative to a small Dice/IoU reduction suggests attention faithfulness and segmentation accuracy are, at least partially, separable objectives: the λ_2 sweep's monotonic AAMO

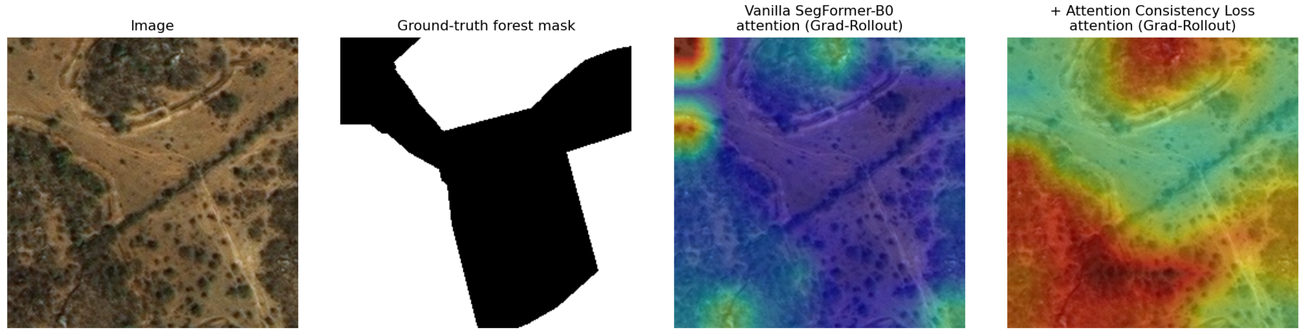

Figure 2: Qualitative comparison on a held-out test image (left to right): input image; ground-truth forest mask; vanilla SegFormer-B0 attention (Grad-Rollout); attention after adding the Attention Consistency Loss (λ_2=0.3 checkpoint; λ_2=1.0 map pending). Full-scale checkpoints (5,108 images, 20 epochs, best-val). The attention-consistency map shifts attention broadly toward the darker canopy clusters and away from the bare ground, consistent with the AAMO gain, but still does not cleanly separate canopy from non-forest by eye. The vanilla model's attention remains dominated by corner/edge hot spots rather than the motivating roads/shadows failure mode even at full scale, indicating this is not simply an undertraining artifact as earlier speculated from the 8-epoch smoke run. Genuine output of the trained checkpoints; one of three qualitative comparisons generated, not otherwise cherry-picked.

trend with a near-flat Dice band indicates a tunable trade-off, not a sharp accuracy collapse. This now holds at full scale (5,108 images, 20 epochs) as well as at the earlier smoke scale, though a multi-seed ablation is still required before drawing firm conclusions. Both the MSE and KL-divergence formulations of L_{att} are implemented; only MSE has been evaluated so far.

Threats to validity. (1) *Scale.* U-Net and both SegFormer-B0 variants are now trained and evaluated on the full 5,108-image dataset for 20 epochs, sharing the same 766-image held-out test set, resolving this threat for those three rows; DeepLabV3+ remains a 200-image smoke-scale reference only, so its comparison to the other rows should be read with that caveat. The SegFormer checkpoints are early-stopped at their best-validation epoch rather than fully converged. (2) *Single seed.* No run has been repeated with a different seed, and the λ_2 sweep that selects λ_2=1.0 is itself single-seed, so the reported numbers carry unknown variance; Phase 2's multi-seed ablation is required before any number here is treated as final. (3) *CNN baseline provenance.* The reported U-Net baseline is now the PyTorch model trained with the identical split, seed, and training loop as the SegFormer runs (resolved from an earlier draft that mixed in a separately-trained Keras model); Dice/IoU are computed dataset-wide (TP/FP/FN accumulated over all 766 test images), consistent with how the other rows in Table 1 are computed. (4) *Rollout restriction.* Restricting Gradient-weighted Attention Rollout to SegFormer's stage-4 attention avoids the invalid rectangular-matrix case but discards information from stages 1–3, which this work does not yet quantify the cost of.

Broader impact. Forest-cover monitoring increasingly informs deforestation-tracking and conservation-policy decisions, where an automated model's *explanation* of a prediction — not just its accuracy — affects whether domain experts and policymakers can trust it. An attention map that reliably attends to canopy rather than roads or shadows is a step toward auditable forest monitoring, but these preliminary, single-seed results are not yet reliable enough to inform real deforestation-monitoring decisions.

7 Conclusion and Future Work

We presented an explainability-guided SegFormer-B0 that supervises internal attention against the ground-truth forest mask via an Attention Consistency Loss, an adaptation of Gradient-weighted Attention Rollout to SegFormer's spatial-reduction attention, and a new quantitative interpretability metric, AAMO. Full-scale results on the 5,108-image dataset confirm the core hypothesis: with λ_2=1.0 (validation-swept), AAMO rises ~22× (0.0334⟶0.7476) for a ~1.7-point Dice cost. Future work will run the complete multi-seed ablation study, compare the MSE and KL-divergence loss formulations, extend the λ_2 sweep to multiple seeds, address the residual overfitting, and, time permitting, add a boundary-refinement module targeting canopy-edge errors.

References

[1] 2022. An Attention-Based U-Net for Detecting Deforestation Within Satellite Sensor Imagery. *ScienceDirect* (2022).
[2] Bianca-Cerasela-Zelia Blaga and Sergiu Nedevschi. 2024. Forest Inspection Dataset for Aerial Semantic Segmentation and Depth Estimation. *arXiv preprint arXiv:2403.06621* (2024).
[3] Hila Chefer, Shir Gur, and Lior Wolf. 2021. Transformer Interpretability Beyond Attention Visualization. *arXiv preprint arXiv:2101.03919* (2021).
[4] Bingzhi Chen et al. 2022. TransAttUnet: Multi-Level Attention-Guided U-Net with Transformer for Medical Image Segmentation. *arXiv preprint arXiv:2107.05274* (2022).
[5] 2026. ForResANeXt: Forest/Non-Forest Segmentation with Aggregated Residual Attention Network in Satellite Imagery. *ScienceDirect* (2026).
[6] 2024. Guiding Attention in End-to-End Driving Models. *arXiv preprint arXiv:2405.00242* (2024).
[7] 2026. Enhancing Cross-Regional Generalization in UAV Forest Segmentation Across Plantation and Natural Forests with Attention-Refined PP-LiteSeg Networks. *Remote Sensing* (2026).
[8] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
[9] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. 2021. SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[10] Yuan Zhang et al. 2023. eX-ViT: A Novel eXplainable Vision Transformer for Weakly Supervised Semantic Segmentation. *Pattern Recognition* (2023).